

Brittle-ductile transition in compliant finishing of HVOF sprayed hard WC-Co coating

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ABSTRACT

WC-Co coating with nanolevel surface finish is required in some special components in aviation, automobile and forming industries. In this study, compliant finishing of WC based coating is carried out using an in-house fabricated wheel based SAG tool. Theory of contact mechanics and the interactions of active abrasive particles with the workpiece surface are considered to develop a material removal rate (MRR) model. The brittle-to-ductile transition of WC-Co coating is investigated during compliant finishing with diamond abrasive pads having different particle size. It is observed that the material is purely removed through ductile mode when 6 μm polishing pad is used and the critical chip thickness for brittle to ductile transition lies in the range of 60–100 nm. Moreover, the lower wheel compression and higher wheel speed can slightly enhance the possibility of ductile machining. A low surface roughness of 21 nm is achieved using multistep compliant finishing operations with decreasing abrasive particle size in a sequence. The residual stress of the finished coating is found to be -78 ± 8 MPa which is almost 28% of the ground surface. The cross-section of the finished coating is analysed and no appreciable sub-surface damage is detected.

1. Introduction

Tungsten carbide-cobalt (WC-Co) coating is one of the most successful and widely used thermally sprayed coatings [1]. In this coating, the hard WC particles are embedded into the relatively soft and ductile Co phase [2]. Owing to these favourable microstructural properties, WC-Co coating has been widely used to enhance the tribological performance of engineering components engaged in various industries such as aerospace, pump, textile, automobile, etc. [3]. In addition to that, WC-Co coating with nanoscale surface roughness is highly required for some special components like cylinder of the aeroplane landing gear, cylinder of internal combustion engine and calendar roll of the printing devices [4].

WC-Co coating is mostly deposited using high velocity oxy-fuel (HVOF) spraying process as it has ability to produce good quality coating (i.e., high bonding strength and low porosity, etc.) [5]. However, the as-sprayed coating exhibits higher surface roughness. Therefore, the coating needs to be suitably post-processed to attain the desired surface finish. Surface grinding is generally employed as a post-processing technique for the cermet coating. Few literatures of surface grinding of the WC based coatings are available. Maiti et al. [6]

conducted surface grinding of WC based coating and measured the residual stress of the ground coating. It was observed that the mechanical properties of the ground coating were improved owing to the existence of high compressive residual stress. Masoumi et al. [7] performed grinding of WC based coating using a diamond grinding wheel. It was reported that the cohesion and adhesion strength of the ground coating was slightly higher than that of as-sprayed coating. Tillmann et al. [8] conducted surface grinding and ball burnishing of the cermet coating in a sequence. It was found that the tribological performance of the coating was considerably enhanced upon finishing. Pishva et al. [9] studied the corrosion behaviours of the as-sprayed and ground WC-Co-Cr coatings by performing potentiodynamic polarisation tests. It was observed that the corrosion resistance of the ground coating was significantly higher than that of the as-sprayed coating. This is attributed to the comparatively lower surface roughness of the coating. In all the cases, the surface roughness of ground coating was in the sub-micron scale.

Few studies related to the precision finishing of sintered tungsten carbides are reported in the literature. Kim and Lee [10] fabricated an in-process electrolytic dressing system coupled with metal bonded grinding wheel to achieve nanoscale surface finish of the sintered carbides. However, this process requires a complex system for performing

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efficient finishing. Jahan et al. [11] conducted die-sinking micro-electro-discharge (EDM) machining of sintered WC-Co composite using silver tungsten (AgW) electrode. It was observed that the nanoscale surface roughness could be obtained with a very low gap voltage and capacitance. However, the events of arcing and short-circuiting were frequently occurred and that leads to the formation of surface or sub-surface damages on the finished surface. Yin et al. [12] conducted precision grinding of tungsten carbide composite to attain high degree of surface finish. However, the rigidity, stiffness and accuracy of the machine tool needed for precision finishing are high and that makes the process uneconomical. Wayne and Sampath [13] compared the microstructural and mechanical properties of sintered WC-Co material and thermally sprayed WC-Co coating. In the case of sintered material, the homogeneous microstructure with almost no porosity and the presence of pure WC and Co phases lead to the superior metallurgical bonding and mechanical properties. On the other hand, the presence of porosity and metastable phases such as W_2C , W and Co_xW_xC compounds in the thermally sprayed coating lead to the poor inter-splat bonding and poor carbide-matrix bonding and that finally result in, a lower fracture toughness and hardness [14]. Therefore, it is difficult to attain nanolevel surface finish on thermally sprayed coating using rigid abrasive tools. In addition to that, the higher cutting force associated with conventional machining process leads to either pull out of the sprayed particles or damage (i.e., burning) the finished coating. This is attributed to the comparatively poor mechanical bonding of the splats and lower heat transfer rate from the finishing zone owing to the presence of inherent porosities in the coating [15]. This situation demands a flexible finishing process for attaining nanolevel surface roughness on the coating.

Shape adaptive grinding (SAG) is an advanced complaint finishing process used over a wide range of materials [16]. The semi-elasticity of the SAG tool conform the local profile of the components which makes it suitable for the finishing of freeform surfaces. Beaucamp et al. [16] conducted SAG of SiC optics and found that material was mostly removed through plastic flow in SAG. The form error was reduced from 40 μm to 112 nm (peak-to-valley). Beaucamp et al. [17] performed SAG of titanium based alloy (Ti6AL4V) produced by selective laser sintering (SLS). A highly irregular layer was found on the outer surface of the as-deposited titanium alloy. It was observed that the outer layer could be successfully removed by SAG and the surface roughness was reduced to around 10 nm (R_a). Zhu and Beaucamp [18] performed the finishing of zerodur and cordierite ceramics using SAG with a diamond abrasive pad having 3 μm particle size. The effects of precess angle and tool offset on the surface roughness and MRR were investigated. An increasing trend of MRR was observed with an increase in tool offset and precess angle, whereas no specific trend was found for the surface roughness. Zhu et al. [19] investigated the material removal mechanism of individual abrasive particle situated at various locations of SAG tool. Based on the experimental results, a spring-grain model was proposed for better understanding of the interactions between the abrasives and workpiece surface. Ghosh et al. [20] performed SAG of oxygen-free high conducting (OFHC) copper with benzotriazole (BTA) solution and achieved nanoscale surface roughness without producing any oxidized film.

Apart from SAG, bonnet polishing is also classified as a complaint finishing process due to the existence of elastic support in the finishing tool. Several studies on the bonnet polishing are previously reported in the literature. Cao et al. [21] studied the surface generation characteristics in bonnet polishing using both the simulation and finishing experiments. It was found that tool offset, precess angle and head speed are the main controlling parameters in bonnet polishing. Wang et al. [22] conducted finite element analysis (FEA) to determine the pressure distribution over the contact area in the bonnet polishing. Based on that, the shape of the finishing spot (i.e., influence function) at different wheel compressions can be evaluated. Pan et al. [23] studied the effect of process parameters on the friction coefficient between the workpiece and bonnet polishing tool. It was observed that the friction coefficient has a significant effect on the influence function. Based on those results,

a model of influence function was proposed. Su et al. [24] investigated the shape preserving capability of bonnet tool using both simulation and experiments. The contact condition was identified as the main influential parameter. It was found that the change in form of the bonnet tool after finishing could be restricted to 20 nm under suitable condition. Zeng and Blunt [25] proposed a material removal rate (MRR) model for the bonnet polishing based on Preston's equation. The relevant process parameters of the bonnet polishing were incorporated in the proposed model to assess the effect of each parameter on MRR. However, a large numbers of experiments are required to determine the Preston coefficient. Bo et al. [26] investigated the influence of abrasive wear on the material removal rate (MRR) in bonnet polishing. It was observed that the shape of influence function and the maximum penetration depth were varied with the increase of abrasive wear. Yang et al. [27] developed a contact model for finishing of thin-walled components using a complaint tool. A method was presented to obtain uniform MRR by controlling the tool offset. The correlations between the actual and nominal tool offsets were also reported. Feng et al. [28] reported that the influence function in bonnet polishing is largely relied on the tangential velocity and pressure exerted on the finishing zone. Moreover, finite element analysis (FEA) was also performed to determine the optimum parameter for better surface finish. Ihara et al. [29] investigated the material removal mechanism at the center region of the finishing tool during bonnet polishing. It was observed that the material was removed through the mechanical abrasion and the finishing rate could be further improved by the chemical reaction between the polishing fluid and workpiece.

In spite of having notable industrial applications of the nanofinished WC-Co coating, precision finishing of the WC-Co coating is rarely addressed in the literature. In the present investigation, shape adaptive grinding (SAG) of WC based coating is carried out using in-house fabricated wheel based SAG tool. The effects of wheel speed, wheel compression, and the abrasive particle size on the finishing forces, MRR and areal surface roughness are studied. A MRR model is proposed based on the theory of contact mechanics and interactions of active abrasive particles with the workpiece surface and the proposed MRR model is experimentally verified. The brittle-to-ductile transition of WC-Co coating in SAG is investigated in detail. The correlations of material removal mechanism with the process parameters are also analysed. The cross-section and residual stress of the finished coating are analysed to study the sub-surface condition. An areal surface roughness (S_a) of 21 nm is attained by performing SAG with decreasing abrasive particle size in a sequence.

2. Experimental procedures

In the present investigation, a commercially available WC-12Co powder (Metco 72F-NS) was used as the coating material. High velocity oxy-fuel (HVOF) (HIPOJET 2700) spraying technique was utilized to deposit the coating on the steel substrates. The spraying parameters are listed in Table 1. The thickness of the deposited coating was in the range of 300–400 μm . The coating properties were measured in previous study and those are listed in Table 2.

Compliant finishing was carried out on the pre-polished coating which was processed by surface grinding. The finishing direction was kept perpendicular to the grinding direction. The finishing tool consists

Table 1
HVOF process parameters [30].

Acetylene pressure	0.34 MPa
Acetylene flowrate	55 lpm
Oxygen pressure	0.54 MPa
Oxygen flowrate	290 lpm
Powder carrier gas pressure	0.20 MPa
Standoff distance	150 mm
Gun travel speed	300 mm/min

Table 2
Properties of WC-Co coating [31].

Coating properties	Values
Microhardness (GPa)	11.02 ± 1.2
Elastic modulus (GPa)	200 ± 21
Fracture toughness (MPa m ^{1/2})	7.78 ± 0.9

of three different layers as schematically shown in Fig. 1 (a). Polymethyl methacrylate (PMMA) wheel is used as the base material and the intermediate layer is made of polyurethane foam (i.e., elastic layer) and diamond polishing pad acts as the outermost layer as shown in Fig. 1 (a). The diameter and width of the finishing wheel are 125 mm and 10 mm, respectively. The experimental set up of compliant finishing is shown in Fig. 1 (b). The compliant finishing tool was attached to a CNC machine spindle. Wheel speed (N), wheel compression (T), and particle size of the diamond polishing pad (d_p) were chosen as the input parameters and those are given in Table 3. Wheel compression is the squeezing of elastic tool from the first contact of the workpiece surface. Three resin bonded diamond polishing pads (UltraPrep, Buehler, Switzerland) having particle sizes of 30 μm , 15 μm and 6 μm were used in the study. During finishing operation, a raster tool path with a feed rate of 15 mm/min was engaged. A load cell (0.1 N resolutions) was used for the measurement of finishing forces (both normal and tangential forces) as shown in Fig. 1 (b).

Surface roughness of the finished surface was measured using a 3D non-contact profiler. The volume of material removed after each spot finishing experiment was also measured using the non-contact profiler. The microstructures and cross-sections of the finished coatings were characterized using a scanning electron microscope (SEM). The cross-section of the finished coating was also characterized using SEM. X-ray diffraction (XRD) technique was used to measure residual stress of the as-ground and the finished coatings. In this investigation, Cu K_α radiation ($\lambda_{\text{Cu}}=1.54 \text{ \AA}$) was employed for the measurements. The reflection from the (201) crystallographic plane of WC phase was analysed to evaluate the lattice strain [31]. During measurement, $\text{Sin}^2\psi$ method was followed by engaging several stage tilt angles (ψ) as follows: 0° , $\pm 26.67^\circ$, $\pm 33.21^\circ$, $\pm 39.23^\circ$, and $\pm 45^\circ$.

3. Modeling of MRR

When the compliant finishing tool is brought in contact with the ground surface, a line contact is initially established as shown in Fig. 2

Table 3
Levels of process parameters.

Process parameters	Levels			
Wheel speed (N) (rpm)	300	550	800	1050
Wheel compression (T) (mm)	0.2		0.4	0.6
Particle size of polishing pad (d_p) (μm)	6		15	30

(a). During finishing operation, the elastic wheel gets pressed against the workpiece and that leads to the elastic deformation of the tool in the vicinity of their first contact. Therefore, the contact is made over an area as schematically shown in Fig. 2 (b).

However, the contact area is comparatively smaller than the dimension of both the contacting bodies. A suitable theory of contact mechanics needs to be applied to predict the approximate shape of the contact area and distribution of the contact pressure. By considering the problem of non-conforming bodies, it can be tentatively assumed that the shape of contact area is elliptical [32,33]. Such an elliptical contact patch with the semi-axes of a , and b is schematically shown in Fig. 2 (b). To confirm the hypothesis, the actual macrograph of the finishing spots are analysed and once such macrograph is shown in Fig. 3. It is observed that the shape of contact area is approximately elliptical. According to the Hertz contact theory, the pressure distribution over the finishing zone can be expressed by Eq. (1) [34].

$$p = p_0 \left\{ 1 - (x/a)^2 - (y/b)^2 \right\}^{1/2} \quad (1)$$

where, p_0 is the maximum pressure at the centre of the finishing spot.

The total contact load (F_N) at the contact region of compliant finishing tool and workpiece can be calculated with the product of average contact pressure (i.e., $p_{av} = 2/3 \cdot p_0$) and the elliptical contact area (i.e., πab). However, it is more convenient to express the contact load (F_N) in terms of mechanical properties of the contacting bodies and elastic deformation of the SAG tool and the same can be expressed by Eq. (2) [33].

$$F_N = 1.44 E_{eq} R^{1/2} T^{3/2} \quad (2)$$

where, E_{eq} is the equivalent elastic modulus of the contact, R is the radius of the finishing wheel and T is the wheel compression.

The equivalent elastic modulus (E_{eq}) can be derived from the elastic moduli (E_w and E_t) and Poisson's ratio (ν_w and ν_t) of the WC-Co coating (i.e., workpiece) and the compliant finishing tool. It can be expressed as

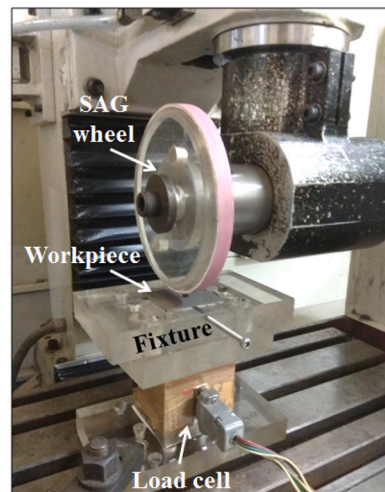
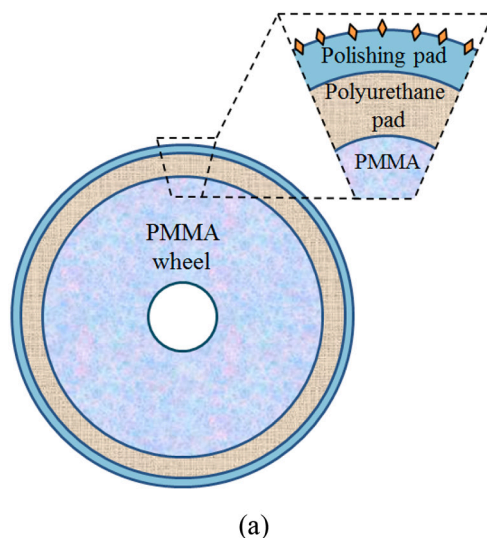


Fig. 1. (a) Schematic of compliant finishing tool showing its constituent layers, and (b) experimental set-up of compliant finishing.

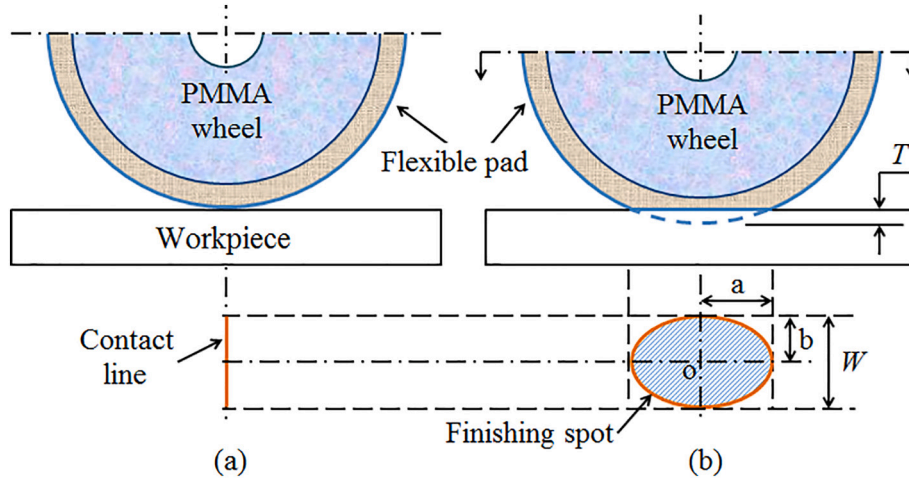


Fig. 2. Schematic of the contact between the compliant finishing tool and workpiece surface during (a) initial contact and (b) finishing operation (i.e., pressed condition).

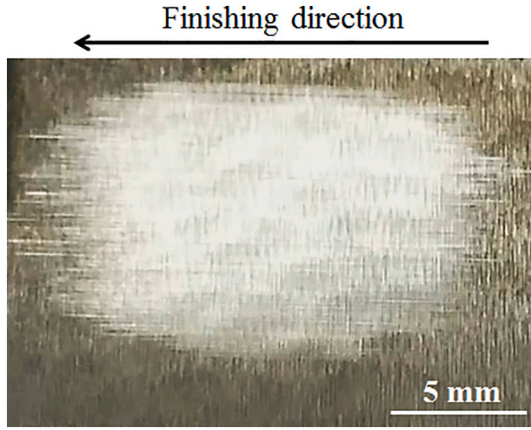


Fig. 3. Macrograph of finishing spot placed on the ground surface.

follows [34]:

$$E_{eq} = \left(\frac{1-\nu_w^2}{E_w} + \frac{1-\nu_t^2}{E_t} \right)^{-1} \quad (3)$$

The elastic modulus of the backing pad (E_t) can be evaluated from its shore hardness (S) as given by Eq. (4) [16].

$$E_t = \frac{0.0981(56 + 7.62S)}{0.1375(254 - 2.54S)} \quad (4)$$

The Poisson's ratio of the backing pad (ν_t) is taken as 0.24 [35], the elastic modulus (E_w) and Poisson's ratio (ν_w) of the workpiece material are taken as 200 GPa and 0.25, respectively [36].

The tangential force (F_T) can be calculated from the friction coefficient (μ) of the ground coating as given by Eq. (5). The friction coefficient (μ) is taken from Ghosh et al. [31].

$$F_T = \mu F_N \quad (5)$$

The normal force (F_N) and tangential force (F_T) are calculated at different tool compressions (T) by Eq. (2) and Eq. (5), respectively. The finishing forces are also experimentally measured using a load cell. In both the case of normal and tangential finishing forces, a good agreement is observed between the experimental and theoretical results as shown in Fig. 4. It is observed that the experimental finishing forces are slightly greater than the theoretical ones. The attachment of diamond abrasive pad on the elastic backing may slightly enhance the rigidity of

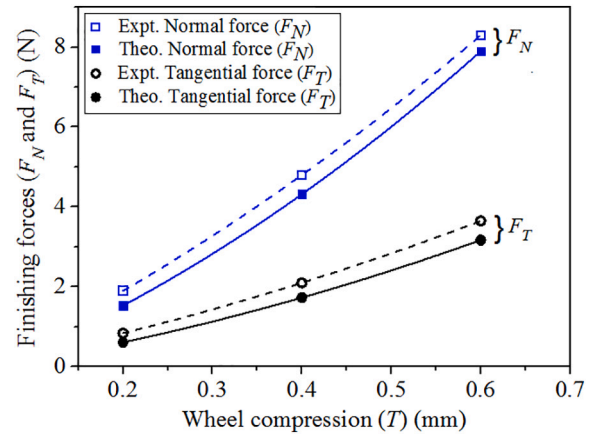


Fig. 4. Comparison of experimental and theoretical finishing forces at different wheel compressions (T).

the compliant finishing tool and that increases the experimental finishing forces.

A set of finishing experiments are conducted with different parameter settings (as listed in Table 2) to analyse the effect of process parameters (i.e., wheel speed (N) and wheel compression (T)) on the finishing spots. All the finishing spots are analysed by the non-contact profiler. The topography of a typical finishing spot is shown in Fig. 5 (a). The areas of all the finishing spots (A_s) (mm^2) are measured and the results are fitted using a polynomial equation ($R^2 = 0.991$) function of the wheel compression (T), and wheel speed (N). It can be expressed by Eq. (6) as follows.

$$A_s = 138.22 (T^{0.151} \cdot N^{0.009}) \quad (6)$$

The cross-section along the length of a finishing spot (i.e., along L-L) is displayed in Fig. 5 (b). The length of all the finishing spots (L_s) (mm) are measured and the results are fitted using a polynomial equation ($R^2 = 0.981$) function of the wheel compression (T), and wheel speed (N). It can be expressed by Eq. (7).

$$L_s = 17.69 (T^{0.232} \cdot N^{0.012}) \quad (7)$$

To determine the number of abrasives present over the spot area, the density of abrasives on the respective abrasive pad needs to be studied. A few SEM micrographs of the polishing pads are analysed to assess the density of the abrasive particles. Such a secondary electron image of the

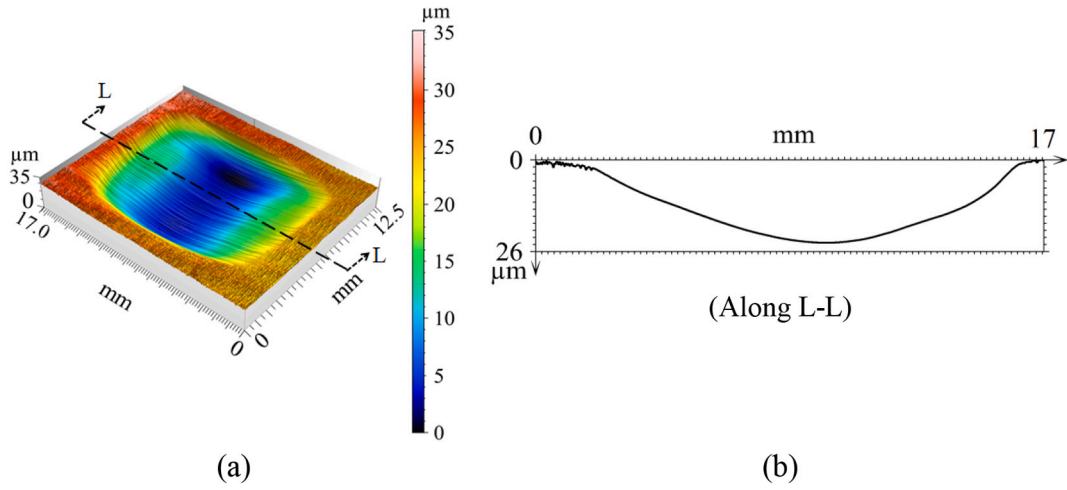


Fig. 5. (a) 3D topography of a finishing spot and (b) 2D surface profile along the length of the finishing spot (i.e., along L-L).

polishing pad of 30 μm particle size is shown in Fig. 6. For 30 μm abrasive pad, the density of the abrasives (C_o) is found as 312 mm^{-2} . Ghosh et al. [4] reported that the density of active abrasive (C_a) during SAG is almost 80 percentage of the abrasive particle density (C_o) (i.e., $C_a = 0.8C_o$). Therefore, total numbers of active abrasive particles (N_{abr}) present over the contact area (i.e., finishing spot) can be evaluated by Eq. (8).

$$N_{abr} = C_a A_s \quad (8)$$

To simplify the analysis, it is presumed that the normal (F_n) and tangential (F_t) forces are uniformly distributed over the active abrasive particles and the abrasives are spherical in shape with identical size. Thus, the normal force (F_n) and tangential force (F_t) exerted on each active abrasive can be evaluated by Eq. (9) and Eq. (10), respectively.

$$F_n = \frac{F_N}{N_{abr}} \quad (9)$$

$$F_t = \frac{F_T}{N_{abr}} \quad (10)$$

To determine the depth of penetration (d) of an abrasive (Fig. 7), the equation of Brinell hardness number (BHN) is used as given by Eq. (11) and Eq. (12).

$$\text{BHN} = \frac{2F_n}{\pi d_g (d_g - \sqrt{d_g^2 - d_i^2})} \quad (11)$$

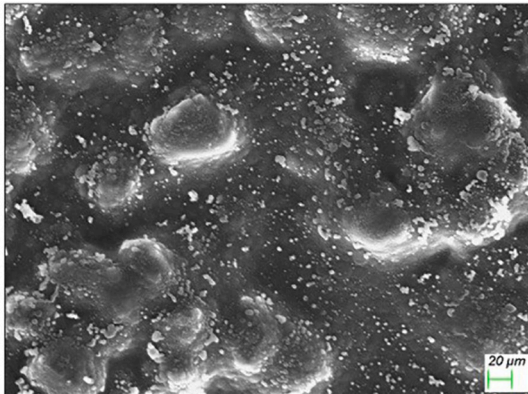


Fig. 6. Secondary electron image of a diamond polishing pad ($d_g = 30 \mu\text{m}$).

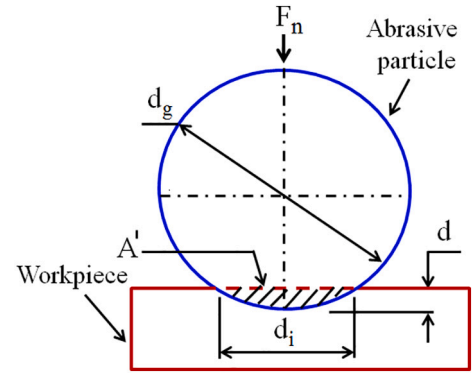


Fig. 7. Schematic of indentation of a spherical abrasive particle into the workpiece.

$$d = \frac{d_g}{2} \left(1 - \sqrt{1 - \frac{d_i^2}{d_g^2}} \right) \quad (12)$$

where, d_g is the mean diameter of abrasives, d_i is the indentation diameter (Fig. 7) and BHN of WC-Co coating is taken as 581 kg/mm^2 [4].

During finishing, the indented abrasive particles move linearly through the finishing zone owing to rotational motion of the compliant finishing wheel. As a result, several linear grooves are formed on the finishing zone. The cross-sectional area of the groove (A') generated by a single abrasive particle can be calculated from the geometry (as shown in Fig. 7) and expressed by Eq. (13).

$$A' = \frac{d_g^2}{4} \sin^{-1} \left(\frac{2 \cdot \sqrt{d(d_g - d)}}{d_g} \right) \sqrt{d(d_g - d)} \left(\frac{d_g}{2} - d \right) \quad (13)$$

Thus, the volume removed by an abrasive (V_a) can be calculated as follows:

$$V_a = L_s \cdot A' \quad (14)$$

The numbers of active abrasives interact with the workpiece surface (n_t) in one rotation can be calculated by Eq. (15).

$$n_t = (\pi DW) C_a \quad (15)$$

where, D and W are the diameter and width of the compliant finishing wheel, respectively.

Finally, the material removal rate (MRR) (mm^3/min) in SAG can be evaluated by Eq. (16).

$$\text{MRR} = V_a \cdot n_t \cdot N \quad (16)$$

where, N is the wheel speed of compliant finishing wheel in rpm.

The 3D topography of all the finishing spots (Fig. 5 (a)) are analysed to measure the volume of material removed after performing compliant finishing with a particular finishing time and thereafter, MRR is calculated. The theoretical MRR in compliant finishing are plotted against the experimental MRR as displayed in Fig. 8. A linear fit ($R^2 = 0.991$) is perceived that indicates a good correlation between the experimental and theoretical results.

4. Results and discussion

To analyse the microstructure of the powder (i.e., WC-12 Co) that was used for the deposition of coating, the powders were cold mounted and polished using 1 μm diamond suspension. Thereafter, the cross-section of the powder was analysed using a scanning electron microscope (SEM) as shown in Fig. 9 (a). It is observed that the WC particles are blocky and angular in shape as shown in inset of Fig. 9 (a). The mean WC grain size is found to be around $1.36 \pm 0.29 \mu\text{m}$. By analysing the high resolution SEM image of powder using Image-J software, it is perceived that the porosity level of the powder was insignificant (0.75%). Secondary electron image of the coating cross-section is shown in Fig. 9 (b). The thickness of the deposited coating is found to be in the range of 300–400 μm . The cross-sectional microstructure of the coating is also shown in Fig. 9 (b). The porosity of the coating is found to be around 1.5%.

The topography of as-deposited coating is shown in Fig. 10 (a). It is observed that several large scale peaks and valleys are randomly distributed over the coated surface. Those topographical features exhibit a higher average protrusion height or valley depth. A surface roughness (S_a) of 5.1 μm is observed on the as-deposited coating and maximum peak-to-valley height (S_z) is found to be around 53 μm .

Several unidirectional grinding marks are observed on the ground surface as presented in Fig. 10 (b). The surface roughness of the ground surface is found to be around 0.6 μm and maximum peak-to-valley height (S_z) is decreased to 6.2 μm . Secondary electron image of the ground coating is shown in Fig. 10 (c). It is observed that material is removed through both brittle fracture and plastic flow. A deformed layer containing pulverized or fractured WC phase and Co binder are found on the top surface of the ground coating as shown in Fig. 10 (d). The inline observations were also reported by several researchers [6,7]. The deformed layer is in highly stressed condition and therefore, a high compressive residual stress of $280 \pm 24 \text{ MPa}$ is found on the ground surface. The stress measurement is performed transverse to the grinding direction. The aggressive mechanical interaction of diamond abrasives with the coated surface promotes the evolution of high compressive

stress on the ground coating [37]. The deformed layer needs to be entirely removed during subsequent finishing operation to restore the desired surface properties of the coating. No sub-surface micro or macro cracks are found on the coating cross-section after surface grinding operation. In the cross-section of ground coating, a very few micro-pores are found. It is believed that those are generated during manual polishing of the coating cross-section using diamond paste with reducing particle sizes of 1 μm and 0.5 μm .

4.1. Effect of process parameters

The contact area in compliant finishing becomes considerably larger than that of conventional grinding due to the presence of elastic backing in the tool. Therefore, more numbers of abrasive grits participate in the finishing operation. As a result, finishing forces acting on each abrasive gets substantially reduced. In the present study, the values of normal force (F_n) and tangential force (F_t) on each abrasive are in the range of 10^{-4} – 10^{-5} N and 10^{-5} – 10^{-6} N , respectively. The low value of normal force (F_n) leads to a very shallow depth of penetration (d) and that results in, nanolevel surface finish on the coated surface.

In the present study, the normal force and tangential force are presented as a function of wheel speed and wheel compression during compliant finishing with 30 μm abrasive pad as shown in Fig. 11 (a) and (b), respectively. With the increase of wheel compression, the elastic deformation of compliant finishing wheel increases and consequently the contact pressure at the wheel-workpiece contact region. Hence, an increasing trend of cutting forces is noticed. On the other hand, it is observed that the wheel speed has a trivial effect on the finishing forces as shown in Fig. 11 (a) and (b). This is attributed to the negligible changes in contact pressure with the variation of wheel speed. Hence, the forces remain almost constant over the entire speed range. The effect of wheel speed and wheel compression on MRR is shown in Fig. 11 (c). With the increase of wheel compression, the finishing forces increase and that leads high penetration depth. Therefore, an increasing trend of MRR is noted with the increase of tool compression. On the other hand, a positive correlation of MRR with wheel speed is also observed though the finishing forces remain almost constant over the entire speed range. This phenomenon can be described based on the Preston's equation that states MRR is proportional to both the contact pressure and relative velocity of the polishing tool [38,39]. Therefore, MRR significantly enhances with the wheel speed.

Surface roughness as a function of wheel speed and wheel compression is shown in Fig. 11(d). At high wheel speed, the number of interactions of active abrasive particles with surface asperities significantly increases in a particular instant. Hence, the protrusion heights of the asperities get reduced in a short time and that results in the considerable reduction in surface roughness. A progressive reduction of surface roughness is observed over the entire speed range when the wheel compression increases from 0.2 mm to 0.4 mm (Fig. 11 (d)). However, at a wheel compression of 0.6 mm, the finishing rate gets substantially retarded beyond a wheel speed of 550 rpm as shown in Fig. 11 (d).

At a wheel speed of 800 rpm, the topographies of finished surfaces corresponding to the wheel compression of 0.2 mm, 0.4 mm and 0.6 mm are shown in Fig. 12 (a), (b) and (c), respectively. During finishing with 0.2 mm wheel compression, the contact pressure is not sufficient to produce the required finishing forces for complete removal of the grinding marks. Therefore, a few residual grinding marks are present over the finished surface and surface roughness of 0.25 μm is achieved as shown in Fig. 12 (a). At a wheel compression of 0.4 mm, the surface roughness gets further reduced to 0.14 μm as shown in Fig. 12 (b). Surface grinding marks are entirely removed and the coated surface gets uniformly polished. When the finishing is performed with 0.6 mm wheel compression, the compliant finishing wheel aggressively interact with the coated surface. As a result, a few deep finishing marks are generated on the coated surface and surface roughness gets comparatively

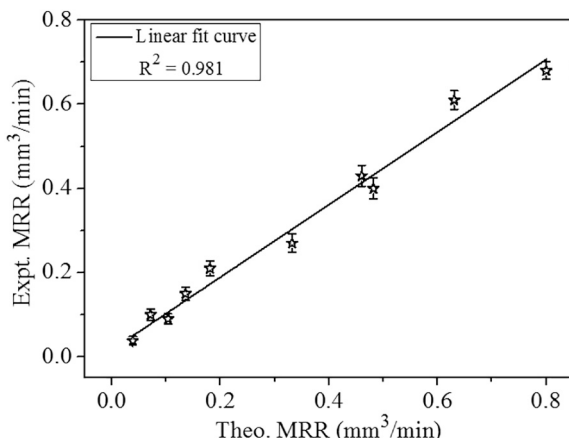


Fig. 8. Comparison of experimental and theoretical results of MRR.

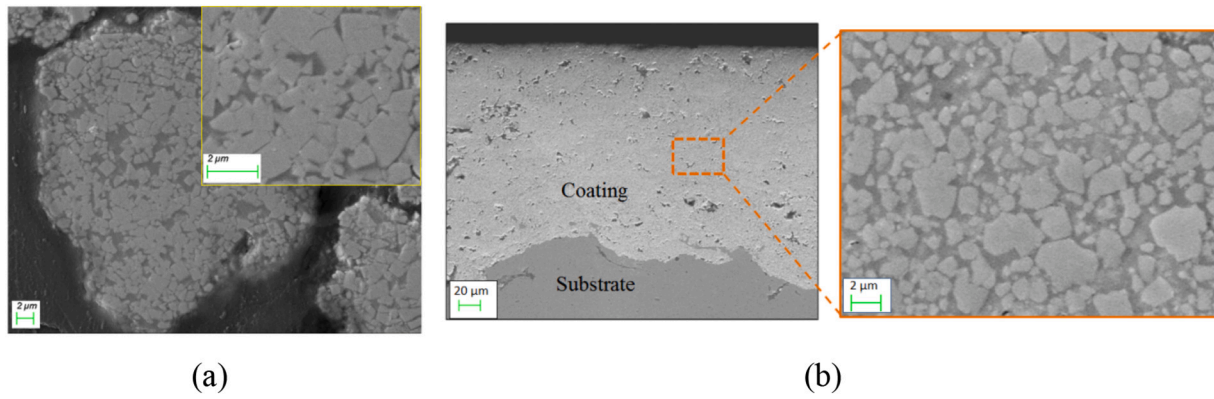


Fig. 9. (a) Cross-sectional microstructure of WC-Co powder, and (b) morphology and cross-sectional microstructure of the coating.

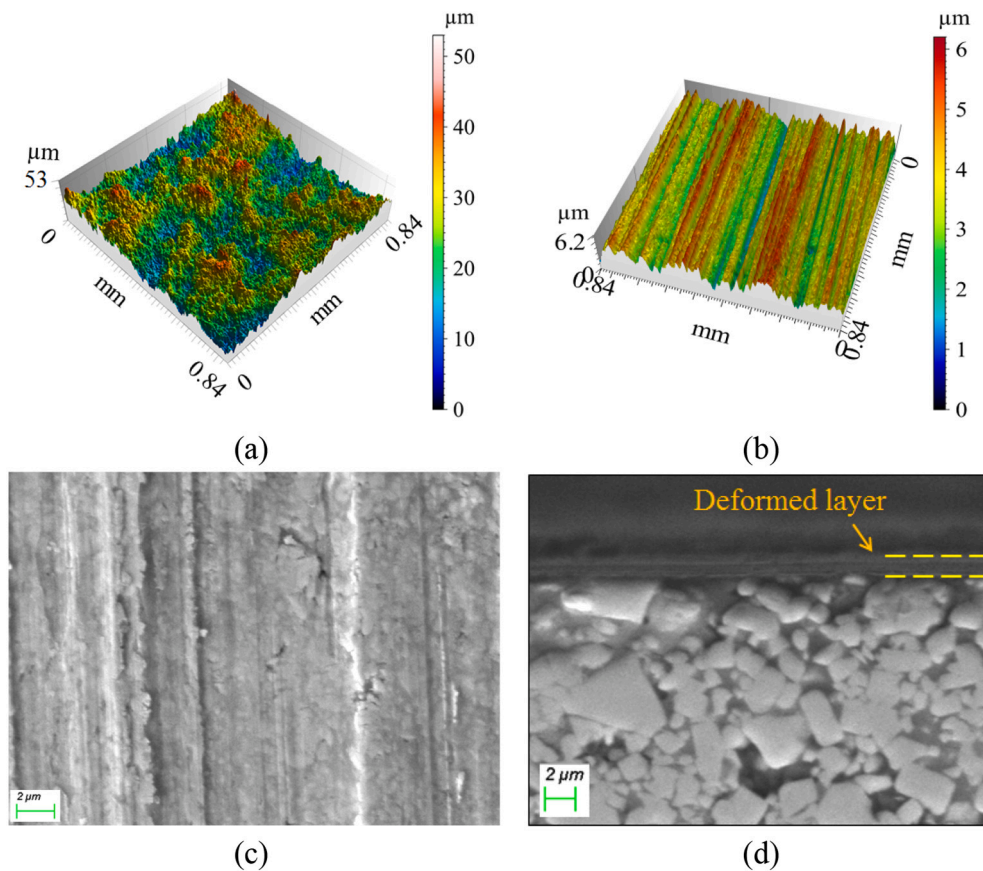


Fig. 10. 3D topography of (a) as-sprayed ($S_a = 5.1 \mu\text{m}$) and (b) ground ($S_a = 0.6 \mu\text{m}$) coatings, and morphology of (c) top surface and (d) cross-section of ground coating.

increased ($S_a = 0.16 \mu\text{m}$) as shown in Fig. 12 (c). From the preceding discussion, it is noted that a medium wheel compression (i.e., $T = 0.4 \text{ mm}$) and higher wheel speed (i.e., $N = 1050 \text{ rpm}$) can produce better surface finish.

To assess the effect of abrasive particle size (d_g) on the response parameters, compliant finishing is performed with the diamond polishing pads having different particle size (i.e., 30, 15, 6 μm). During these experiments, wheel compression is kept as 0.4 mm (i.e., medium level). It is observed that the finishing forces (both normal and tangential) slightly increase with the increase of abrasive particle size as displayed in Fig. 13 (a) and (b). The morphology of the polishing pads needs to be characterized for better understanding of the result. Secondary electron images of the fresh polishing pads of 6 μm , 15 μm and

30 μm grit sizes are displayed in Fig. 14 (a1), (b1) and (c1), respectively. By analysing the pad morphology, the density of the abrasive particles is found to be 1750 mm^{-2} , 720 mm^{-2} and 312 mm^{-2} , respectively. The protrusion height of the surface peaks considerably increases with the abrasive particle size [4]. Thus, the contact load is shared by fewer abrasives in the case of coarse polishing pad. As a consequence, the average indentation depth gets significantly increased with the increase of abrasive particle sizes [40,41]. A higher finishing force is required for the material removal with the same depth of penetration. Hence, a slightly higher finishing force is observed with polishing pads having larger particle size as shown in Fig. 13 (a) and (b).

Furthermore, the volume of material removed by an abrasive particle increases with the indentation depth and that leads to high MRR as

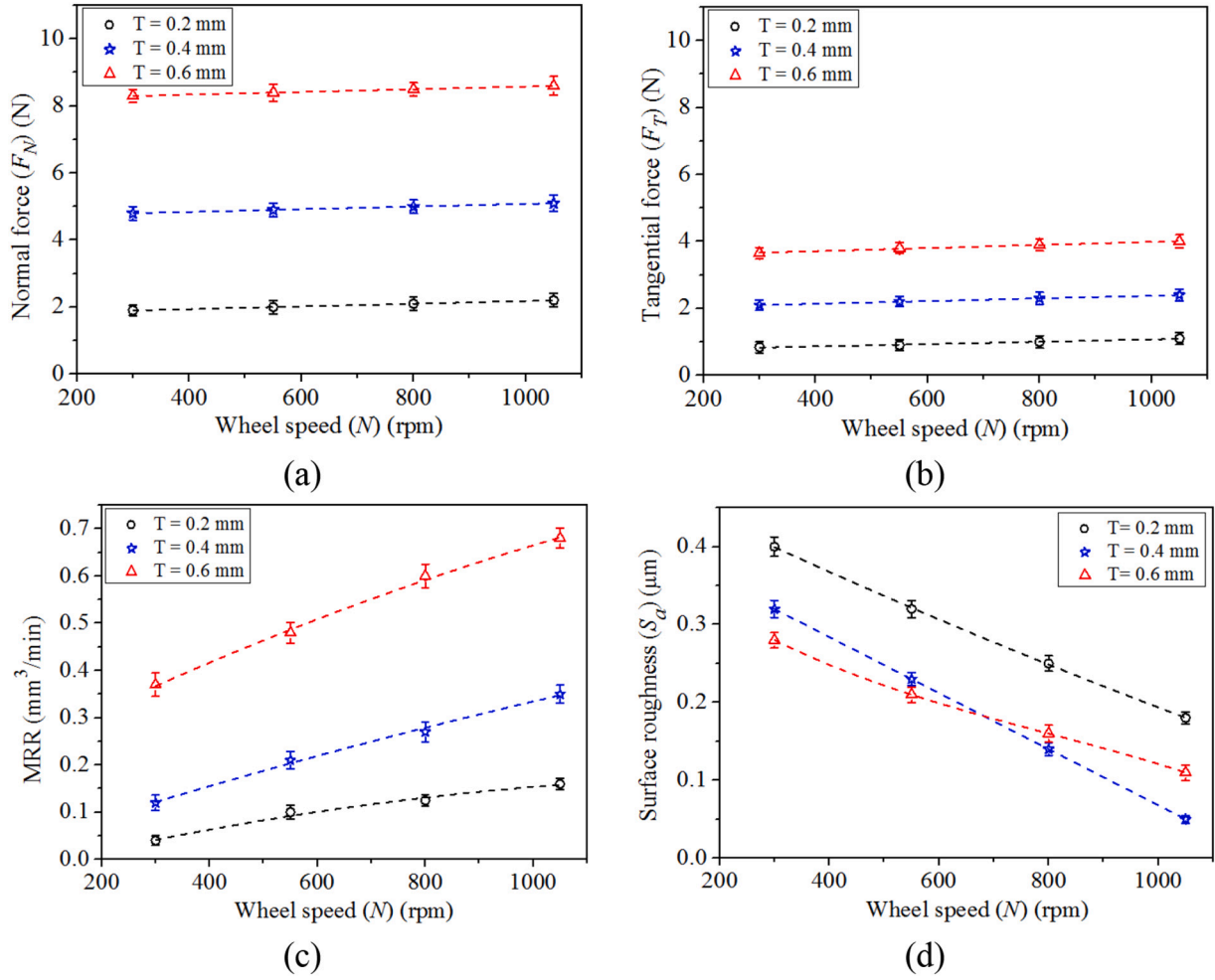


Fig. 11. Variations of (a) normal force (F_N), (b) tangential force (F_T), (c) material removal rate (MRR) and (d) surface roughness (S_a) (with $d_g = 30 \mu\text{m}$) with wheel speed (N) and wheel compression (T).

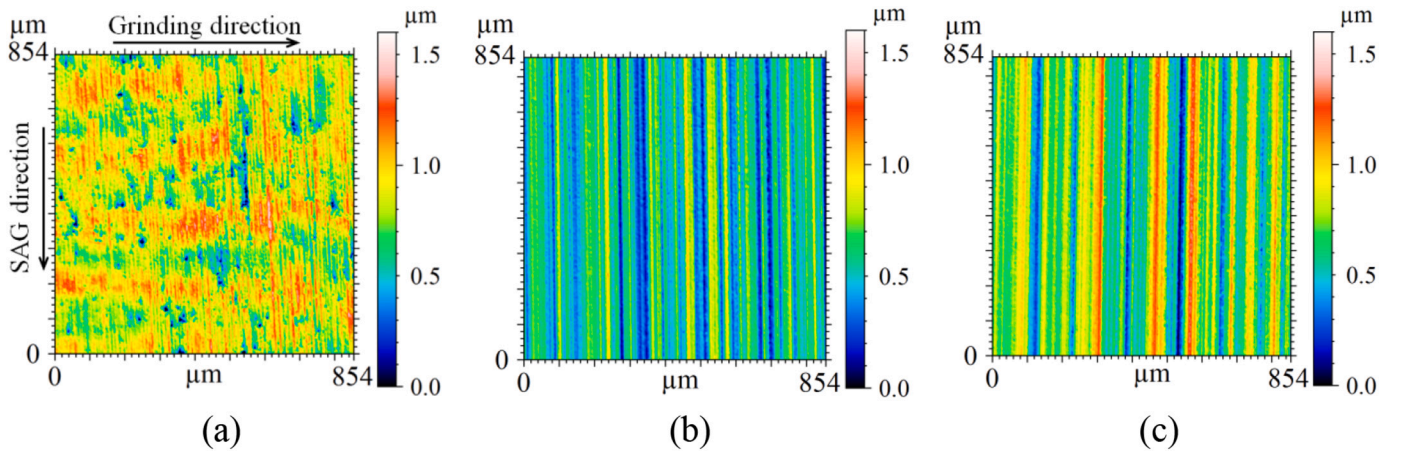


Fig. 12. Pseudo-colour view of the finished surfaces ($d_g = 30 \mu\text{m}$ and $N = 800$ rpm) at a wheel compression (T) of (a) 0.2 mm ($S_a = 0.25 \mu\text{m}$), (b) 0.4 mm ($S_a = 0.14 \mu\text{m}$) and (c) 0.6 mm ($S_a = 0.16 \mu\text{m}$).

shown in Fig. 13 (c). However, the rate of increase of MRR decreases beyond a wheel speeds of 800 rpm in the case of 15 μm and 6 μm polishing pads as shown in Fig. 13 (c). On the other hand, MRR increases steadily over the entire speed range in the case of 30 μm pad. For better understanding, secondary electron images of the used ($N = 1050$ rpm and $T = 0.4$ mm) polishing pads having particle sizes of 6 μm , 15 μm and

30 μm are shown in Fig. 14 (a2), (b2) and (c2), respectively. It is observed that the pad loading is more prominent in the case of 6 μm and 15 μm polishing pads owing to the lower inter-grit spaces for chip accommodation. Therefore, the effectiveness of these pads gets reduced at a higher wheel speed (Fig. 13 (c)). The effect of abrasive particle size on the surface roughness of finished surface is shown in Fig. 13 (d). During

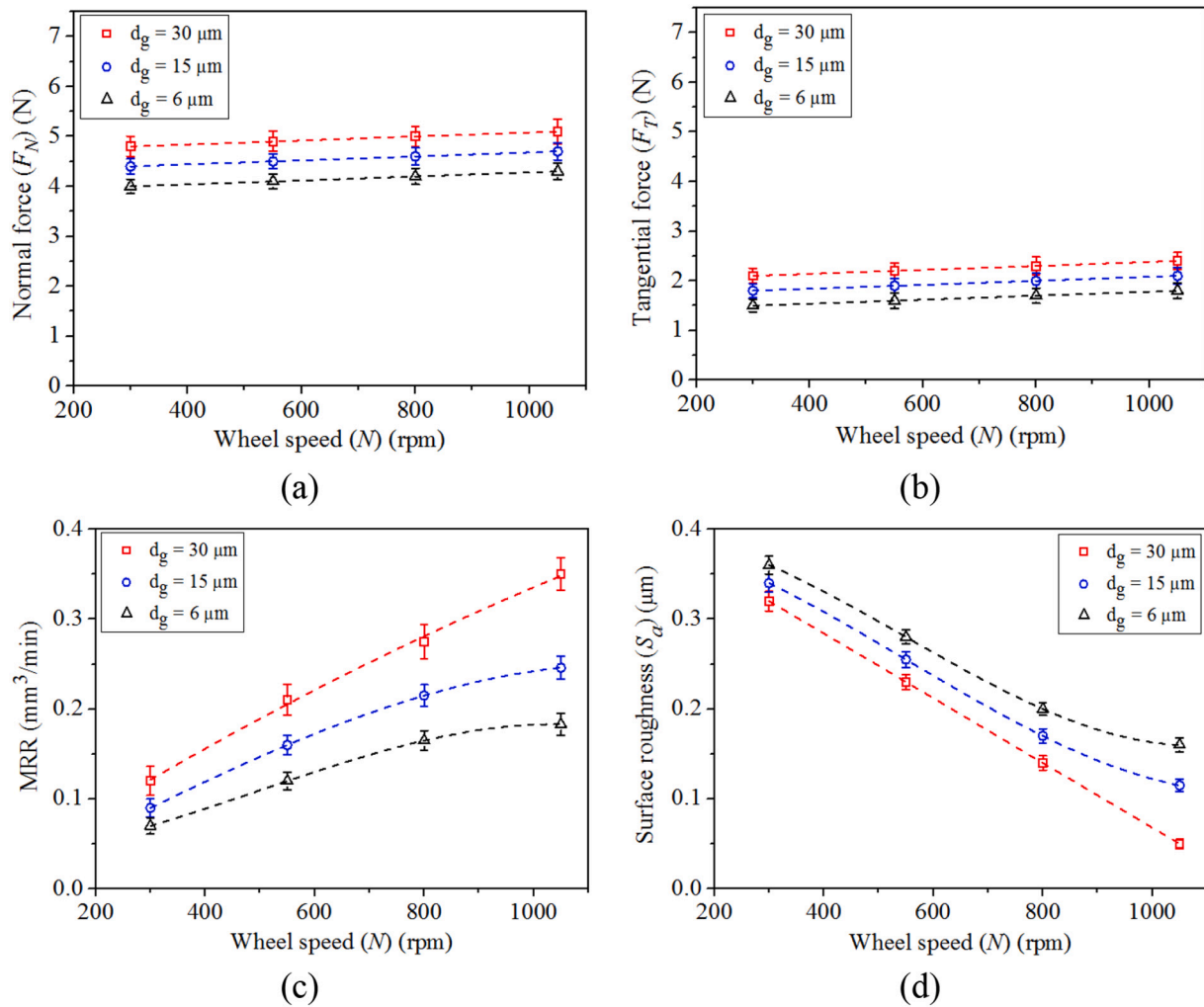


Fig. 13. Variations of (a) normal force (F_N), (b) tangential force (F_T), (c) material removal rate (MRR) and (d) surface roughness (S_a) (at $T = 0.4 \text{ mm}$) with abrasive particle sizes (d_g).

compliant finishing with big particle size, the average height of the surface peaks gets reduced quickly owing to the higher indentation depth and that leads to comparatively lower surface roughness.

The topographies of the finished surfaces processed by compliant finishing ($N = 1050 \text{ rpm}$ and $T = 0.4 \text{ mm}$) with $6 \mu\text{m}$, $15 \mu\text{m}$ and $30 \mu\text{m}$ polishing pads are shown in Fig. 15 (a), (b) and (c), respectively. In the case of $6 \mu\text{m}$ abrasive pad, a few residual grinding marks are found on the finished surface as displayed in Fig. 15 (a). It indicates that the material removal is not sufficient to remove the deep irregularities produced during grinding process. In this case, the surface roughness is reduced to $0.16 \mu\text{m}$. The residual grinding marks are also found on the finished surface processed with $15 \mu\text{m}$ pad (Fig. 15 (b)). In this case, a comparatively lower surface roughness ($S_a = 0.12 \mu\text{m}$) is obtained. During finishing with $30 \mu\text{m}$ abrasive pad, all the irregularities are removed and the surface gets uniformly polished as shown in Fig. 15 (c). A low surface roughness of 50 nm is achieved in this case. Therefore, it is perceived that compliant finishing with the abrasive pad having $30 \mu\text{m}$ particle size is beneficial for the initial finishing of ground coating to obtain a better surface finish.

4.2. Brittle-ductile transition of cermet coating

To assess the material removal mechanism precisely, spot finishing experiments are conducted upto 10 s using different polishing pads ($d_g = 30 \mu\text{m}$, $15 \mu\text{m}$ and $6 \mu\text{m}$) and a specific parameter setting as listed in

Table 3. The fresh polishing pad is used in each experiment to minimize the effect of pad loading. The density of active abrasive particles (C_a) for the abrasive pads having grit size (d_g) of $30 \mu\text{m}$, $15 \mu\text{m}$ and $6 \mu\text{m}$ are found as 250 mm^{-2} , 576 mm^{-2} and 1400 mm^{-2} , respectively. During compliant finishing with coarser abrasive pad, comparatively higher finishing forces are observed (Fig. 13 (a) and (b)). This indicates that the force acting on an active abrasive (i.e., abrasive load) increases with an increase in the particle sizes of the polishing pad. When the indentation depth is substantially small, the energy required for plastic yielding becomes considerably lower than that of brittle fracture and that results in the removal of material through the plastic deformation [42,43].

Secondary electron images of the finished coating processed by compliant finishing with $30 \mu\text{m}$ abrasive pad under different wheel speeds and wheel compressions are shown in Fig. 16 ((a1)-(a3)). Although, the material is mostly removed through plastic flow, several traces of brittle fracture are observed on the finished coating (Fig. 16 ((a1)-(a3))). The characteristics of the finished surface corresponding to the wheel compression (T) of 0.6 mm and 0.2 mm (at $N = 550 \text{ rpm}$) are displayed in Fig. 16 (a1) and (a2), respectively. It is noted that the amount of surface damages (i.e., fractures) decreases with the reduction of wheel compression. As the wheel compression decreases, the contact pressure reduces and that results in, the lower abrasive load and penetration depth [43]. Hence, brittle fracture decreases with the reduction of wheel compression. Surface of the finished coating processed by compliant finishing (at $T = 0.2 \text{ mm}$) with the wheel speeds (N) of 550

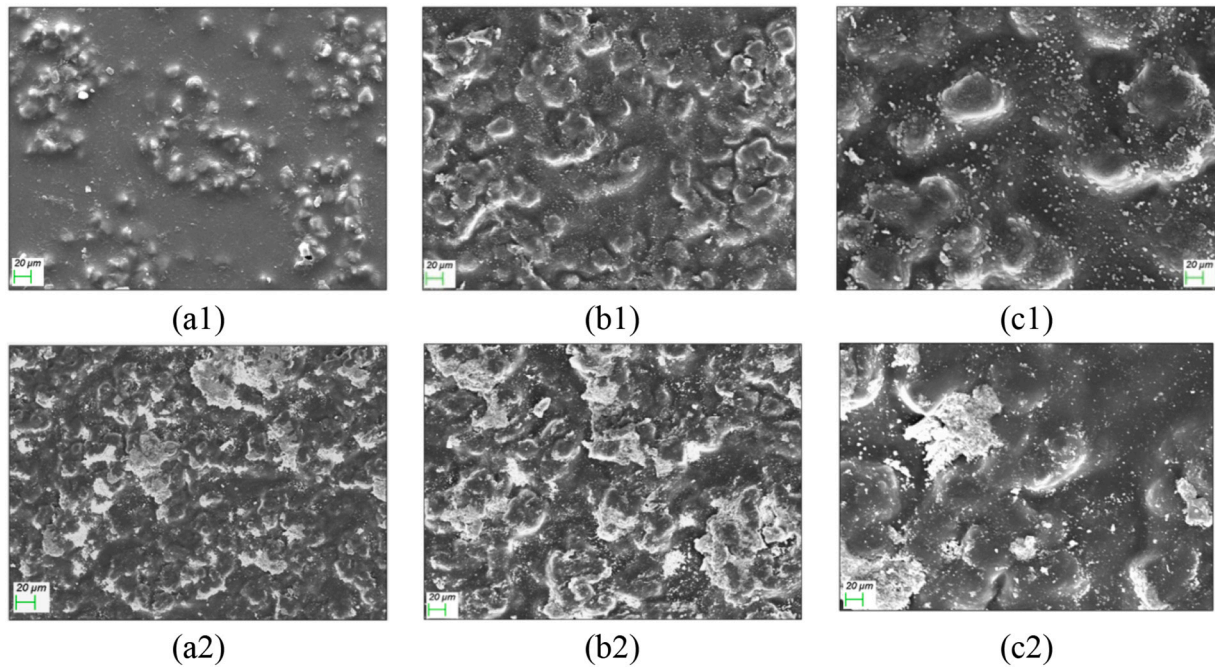


Fig. 14. Secondary electron images of fresh and used polishing pads having particle size (d_g) of ((a1) – (a2)) 6 μm , ((b1) – (b2)) 15 μm , and ((c1) – (c2)) 30 μm .

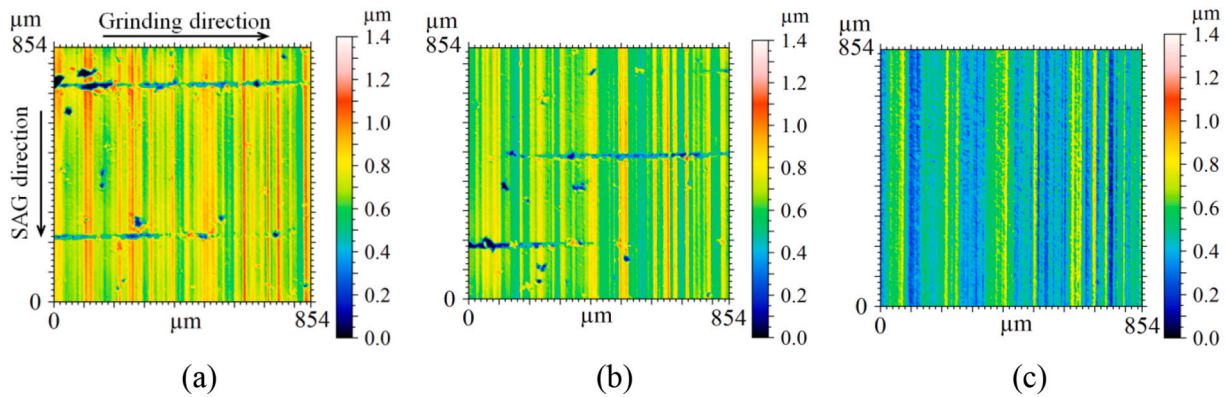


Fig. 15. Pseudo-colour view of finished coating processed by compliant finishing ($N = 1050$ rpm and $T = 0.4$ mm) with abrasive particle size (d_g) of (a) 6 μm ($S_a = 0.16$ μm), (b) 15 μm ($S_a = 0.12$ μm) and (c) 30 μm ($S_a = 0.05$ μm).

rpm and 1050 rpm are displayed in Fig. 16 (a2) and (a3), respectively. It is noted that the traces of brittle fracture decrease with an increase of the wheel speed. Similar observation is also found for the surface grinding operation [44]. In the present study, a slight increase in spot area (A_s) is noted with the increase of wheel speed (Eq. 6) whereas, the finishing forces remain almost constant over the entire speed range (Fig. 11 (a) and (b)). Thus, the same contact load is shared by more numbers of active abrasive particles. Therefore, the abrasive load or penetration depth decreases with the enhancement of wheel speed and that maximize the possibility of plastic deformation. Therefore, surface damages get reduced slightly with the increase of wheel speed.

The microstructures of the finished coating processed with 15 μm polishing pad are shown in Fig. 16 ((b1)-(b3)). In addition to the plastic deformation, a few traces of brittle fracture are identified on the finished coating. However, the amounts of brittle fracture are significantly lower than that of 30 μm pad. This is attributed to the lower contact load and penetration depth in the case smaller particle size. It is observed that the amount of surface damages slightly reduces with the reduction of wheel compression and the increase of wheel speed. This observation is similar to the earlier case as explained previously. Secondary electron images of

the finished surface processed with 6 μm abrasive pad are shown in Fig. 16 ((c1)-(c3)). No brittle fracture is found in this case and the material is removed through pure ductile mode. It is observed that the prominence of ductile mode machining gets further improved with the reduction of wheel compression and the increase of wheel speed (Fig. 16 ((c1)-(c3)). Therefore, it is perceived that the pure ductile mode machining of multiphase brittle materials such as WC-Co coating can be performed using compliant finishing with finer diamond polishing pad. In addition to that, the lower wheel compression and higher wheel speed can slightly increase the possibility of ductile mode machining in compliant finishing irrespective of the particle size of the polishing pad.

In the present study, a single type of WC-Co coating was considered and the mean carbide size (D_{WC}) of the coating was found around 1.36 ± 0.29 μm . The particle sizes (d_g) of the polishing pads under investigation are 30 μm , 15 μm and 6 μm . The ratio ($k = D_{WC} / d_g$) of abrasive particle size to carbide size are found as around 22, 11, and 4. In the cases of SAG with 30 μm and 15 μm pads (i.e., $k = 22$ and 11), several traces of brittle fractures were identified in addition to plastic flow whereas, pure ductile mode machining was observed in the case of 6 μm polishing pad (i.e., $k = 4$). Therefore, it can be conceived that the ratio ($k = D_{WC} / d_g$) of

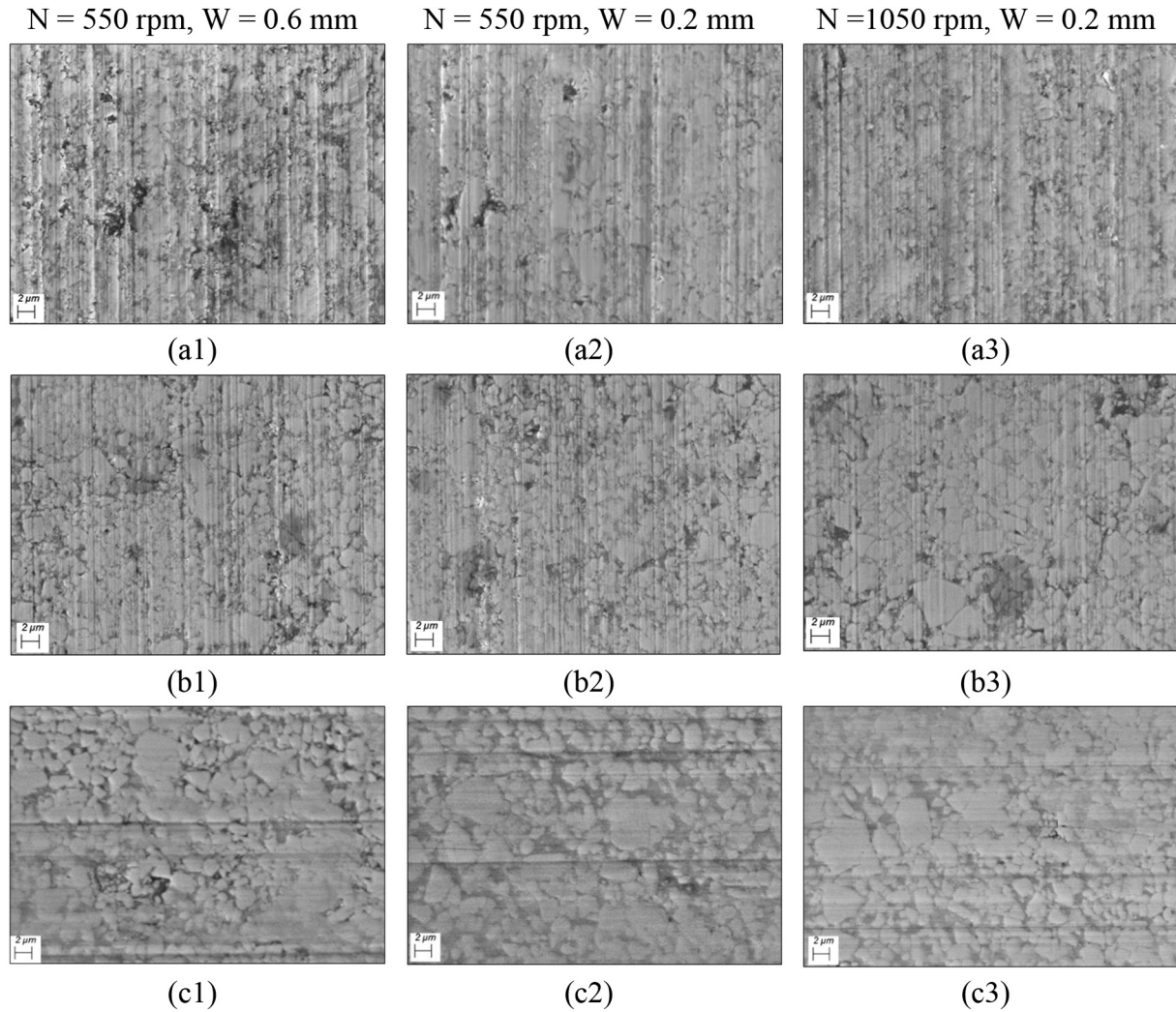


Fig. 16. Microstructures of the finished coatings processed by compliant finishing with diamond polishing pads having grit size of ((a1)-(a3)) 30 μm , ((b1)-(b3)) 15 μm and ((c1)-(c3)) 6 μm .

abrasive particle size to the carbide size needs to be less than five in order to attain pure ductile mode finishing.

In finishing, the ratio of energy required for plastic deformation (G_p) to energy required for brittle fracture (G_f) can be expressed as follows [45,46]:

$$\frac{G_p}{G_f} \sim \frac{f(d^3, d_g, F_n)}{f(d^2, d_g, F_n)} \sim f(d, d_g, F_n) \quad (17)$$

where, d , d_g and F_n are the chip size or penetration depth, abrasive

particle diameter and normal force acting on a single abrasive particle, respectively.

This indicates that the material removal mechanism in finishing is largely relied on the chip thickness, abrasive particle size and abrasive load. The abrasive load (F_n) has positive correlation with the particle size of the polishing pad (d_g). Moreover, it is well documented in the literature that the chip thickness (d) in grinding is inversely proportional with the square root of abrasive particle size (d_g). Therefore, it is perceived that the chips formed during finishing operation can further

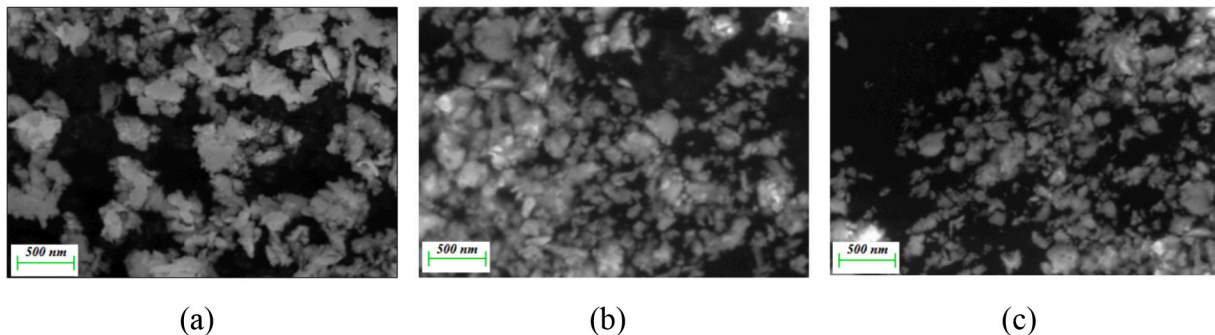


Fig. 17. Secondary electron images of chips produced during compliant finishing with diamond polishing pad having grit size of (a) 30 μm , (b) 15 μm and (c) 6 μm .

shed light on the removal mechanisms. Secondary electron images of chips produced during compliant finishing with abrasive pad having grit size of 30 μm , 15 μm and 6 μm are shown in Fig. 17 (a), (b) and (c), respectively. It is noted that average chip size decreases significantly with the reduction of abrasive particle size. In the case of 30 μm and 15 μm pads, the average chip size is in the range of 240–350 nm and 160–230 nm, respectively. In both the cases, several events of brittle fracture are identified on the finished surface along with large scale plastic deformation. In the case of 6 μm pad, the average size of chip is found in the range of 60–100 nm and material is removed through pure ductile mode. Therefore, it can be perceived that the critical chip thickness for brittle to ductile transition (d_c) for WC-Co coating lies approximately in the range of 60–100 nm.

In the case of brittle material, the critical depth of indentation (d_c) for brittle to ductile transition has been calculated using Bifano's model as given by Eq. (18) [47].

$$d_c = 0.15 \left(\frac{E}{H} \right) \left(\frac{K_c}{H} \right)^2 \quad (18)$$

where, E , H , and K_c are the elastic modulus, Vickers hardness and fracture toughness of the material.

Those properties of WC-Co coating are found as 200 ± 21 GPa, 11.02 ± 1.2 GPa, and 7.78 ± 0.9 MPa $\text{m}^{1/2}$, respectively [31]. Using the Bifano's model (i.e., Eq. (18)), the critical depth of indentation (d_c) for the WC-Co coating is evaluated as 1.37 μm which is almost same as the average grain size of the WC particles (i.e., 1.36 μm [31]). In addition to that, its value is almost 12 times greater than the experimental value of critical chip thickness. This indicates that the Bifano's model is not suitable to determine critical depth of indentation for WC-Co coating. The probable reasons are as follows: (a) mechanical properties of the thermally sprayed WC-Co coatings differ considerably from its sintered or composite counterpart owing to the mechanical bonding among the particles, the presence of porosities, possibility of decarburization, etc., (b) WC-Co coating is multiphase material where the hard and brittle WC particles are embedded into the relatively soft and ductile Co phase, (c) WC grain may contains either prismatic or basal plane which has different nanohardness [13,48,49]. Hence, further study is needed to develop a model to determine the critical penetration depth for WC-Co coating in future.

4.3. Final finishing experiments

From the previous discussion, it is perceived that compliant finishing using 30 μm pad with a wheel compression of 0.4 mm and a wheel speed of 1050 rpm lead to better surface finish. Therefore, the initial step of the final finishing experiment is conducted with this parameters setting. An areal surface roughness (S_a) of 50 nm is attained as shown in Fig. 18 (a).

However, a few deep finishing marks are observed on the polished surface (Fig. 18 (a)). Therefore, in the next step, the finished surface is further polished with 15 μm pad to remove those surface damages. Thereafter, the surface roughness (S_a) is reduced to 32 nm as shown in Fig. 18 (b).

It is previously found that pure ductile mode machining can be attained using compliant finishing with 6 μm pad. Therefore, in the last step, compliant finishing is performed with 6 μm pad for performing finishing in ductile mode. It is observed that the surface is uniformly polished without any deep finishing marks and the surface roughness (S_a) is decreased to 21 nm as shown in Fig. 18 (c). The final finished surface ($S_a = 21$ nm) is further characterized using an atomic force microscope (AFM) as shown in Fig. 19 (a). A few very fine microcutting marks are observed on the surface. It is also observed that the geometrical dimensions of the peaks and valleys are in nanometer scale. A typical 2D surface profile extracted from the topography is shown in Fig. 19 (b). This indicates that the peaks and valleys over the measured area are varied within a range of 16 nm.

Temperature of the finished surface is measured using a thermographic camera immediately after the completion of experiments as shown in Fig. 19 (c). A maximum temperature of 45 $^{\circ}\text{C}$ is observed whereas the room temperature is maintained at 21 $^{\circ}\text{C}$. Therefore, it can be perceived that the possibility of inducing thermal damage on the finished surface is insignificant. Secondary electron image of the cross-section of the finished surface is shown in Fig. 19 (d). The image is taken at an angle to show the top surface and cross-section together. The magnified view of the top surface confirms the presence of fine microcutting marks without any appreciable surface damages. The microstructure of the cross-section is analysed and no appreciable sub-surface damage is observed. The residual stress of the finished coating is found to be around -78 ± 8 MPa which is almost 28% of the ground surface. The stressed layer of the ground coating is entirely removed during compliant finishing. Hence, the residual stress of the finished surface is substantially reduced.

5. Conclusions

In this paper, nanofinishing of WC-Co coating is performed using wheel based compliant finishing tool. A theoretical model of MRR is proposed and brittle to ductile transition in compliant finishing is investigated. The major outcomes of the present study are listed below.

- Due to elastic deformation of compliant finishing tool, number of active abrasive particles increase and the abrasive load decreases significantly. Thus, the penetration depth reduces and finally, nanolevel surface roughness is attained.
- With the increase of wheel compression, the contact pressure, finishing force and MRR increase. However, the deep finishing marks

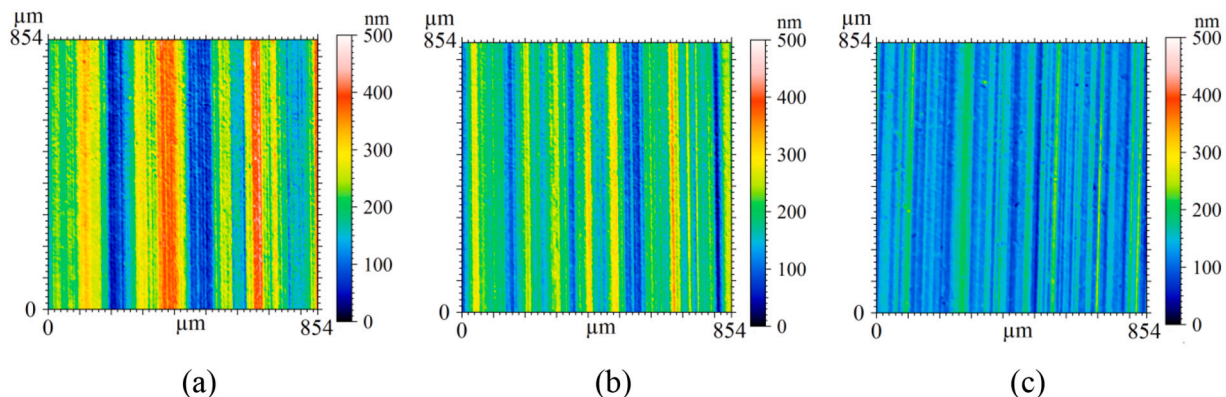


Fig. 18. Pseudo-colour view of the finished coating processed by compliant finishing ($N = 1000$ rpm and $T = 0.4$ mm) with diamond polishing pads having particle sizes (d_g) of (a) 30 μm ($S_a = 50$ nm), (b) 15 μm ($S_a = 32$ nm), and (c) 6 μm ($S_a = 21$ nm) in a sequence.

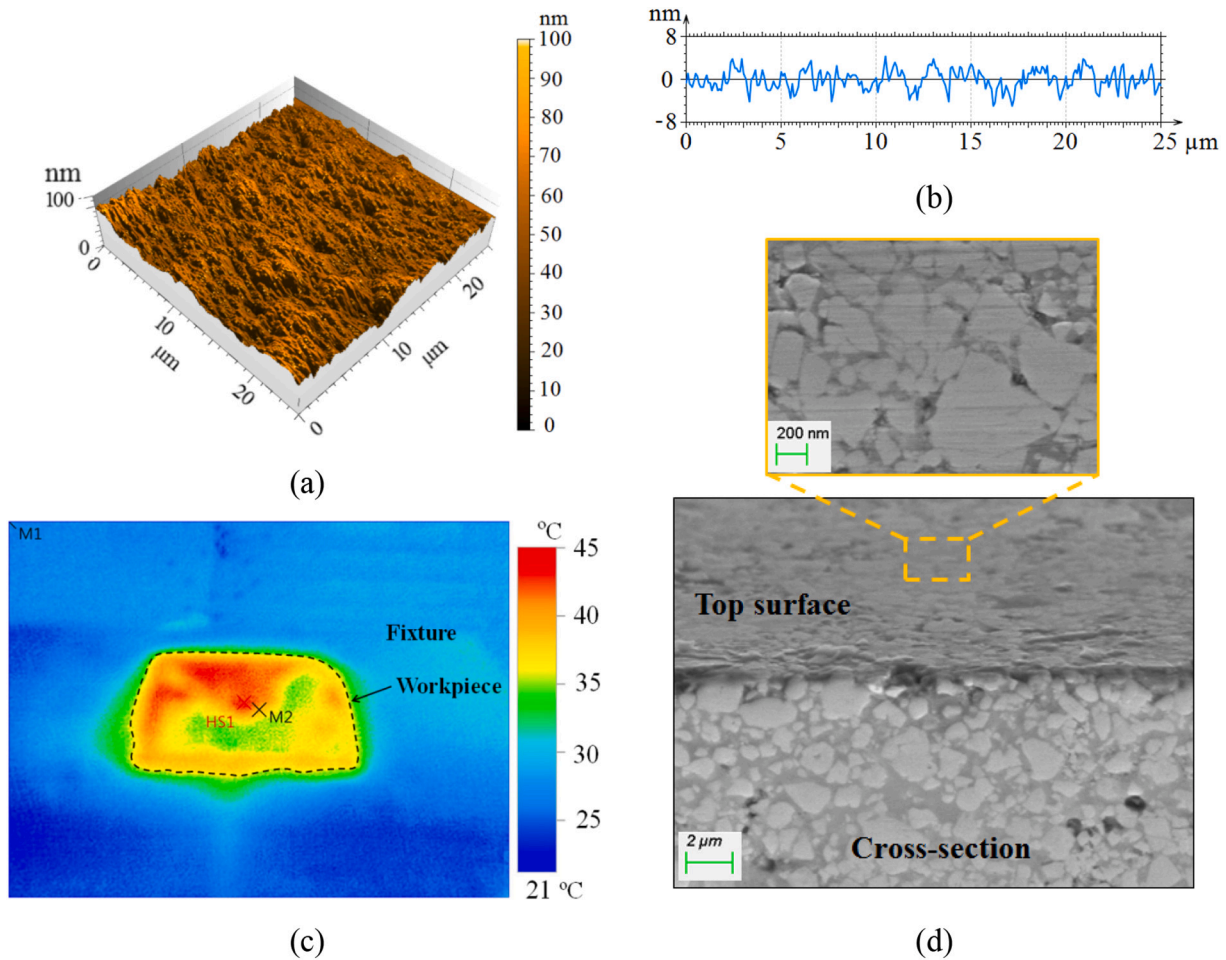


Fig. 19. (a) AFM image showing 3D topography, (b) 2D surface profile, (c) thermographic image and (d) microstructure of top surface and cross-section of the finished surface.

are generated on the surface at a higher wheel compression (i.e., 0.6 mm) and that deteriorates the surface finish.

- The effect of wheel speed on the contact pressure is trivial and that results in, almost constant finishing forces over the entire speed range. However, the numbers of interactions of active abrasives with the surface peaks significantly increase with the enhancement of wheel speed and that leads to the reduction of surface roughness.
- The theory of contact mechanics for non-conforming bodies can be used to determine the contact load in compliant finishing. A theoretical MRR model is developed by considering the Hertz solution and indentation of active abrasive particles. The proposed model is experimentally verified and a good agreement is observed between the experimental and theoretical values.
- The abrasive load, depth of penetration and abrasive particle size are the main controlling parameters for brittle to ductile transition. The abrasive load decreases with reduction of wheel compression and increase of wheel speed. The lower wheel compression and higher wheel speed can slightly increase the possibility of ductile mode machining in compliant finishing irrespective of the particle size of the polishing pad.
- In the case of 30 μm abrasive pad, several traces of brittle fracture are identified on the finished surface along with plastic deformation. The amount of surface damages comparatively reduces during compliant finishing with 15 μm . In the case of 6 μm pad, the material is purely removed through ductile mode. The average chip size is found in the range of 60–100 nm. Therefore, it is perceived that the critical chip

thickness for brittle to ductile transition (d_c) for WC-Co coating lies in the range of 60–100 nm.

- Bifano's model is not suitable to determine the critical depth of indentation for thermally sprayed WC-Co coating owing to the presence of hard and brittle WC particles into the relatively soft and ductile Co phase and considerable difference in mechanical properties between the thermally sprayed coating and its sintered/composite counterpart.
- By performing multistep compliant finishing with decreasing particle sizes in a sequence, a surface roughness of 21 nm is attained. The residual stress of the finished coating is decreased to -78 ± 8 MPa which is almost 28% of the ground surface. A maximum surface temperature of 45 $^{\circ}\text{C}$ is observed on the finished surface and that indicates the possibility of thermal damage is insignificant.

Authors statement

This paper has neither been submitted nor published in any other journal or conference proceedings. This manuscript will not be submitted elsewhere before a decision is made by this journal.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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